Sundry Asset Coverage and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Sundry Asset Coverage (SAC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on SAC achieves an annualized gross (net) Sharpe ratio of 0.32 (0.28), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 22 (20) bps/month with a t-statistic of 3.23 (2.91), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Book to market, original (Stattman 1980), Book leverage (annual), Operating leverage, Equity Duration, Enterprise Multiple, Earnings-to-Price Ratio) is 18 bps/month with a t-statistic of 2.63.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between consumption today and consumption in future states of nature, with risk premia arising from the covariance between asset payoffs and stochastic discount factors (Cochrane and Piazzesi, 2005). While consumption-based asset pricing models provide the theoretical foundation for understanding cross-sectional variation in expected returns, empirical tests often struggle to identify firm characteristics that reliably capture exposure to systematic consumption risk (Lettau and Ludvigson, 2001). We examine whether Sundry Asset Coverage (SAC), defined as the ratio of sundry assets to total assets, predicts stock returns through its connection to firms' sensitivity to aggregate consumption fluctuations and state-dependent marginal utility.

Firms with higher sundry asset coverage maintain larger buffers of miscellaneous and non-operating assets that provide flexibility during economic downturns when marginal utility of consumption is high (Campbell and Cochrane, 1999). These assets, which include prepaid expenses, deferred charges, and other non-core holdings, represent resources that can be liquidated or deployed when external financing becomes costly during bad states of the world. The consumption-based asset pricing framework suggests that assets providing insurance against consumption declines command lower risk premia because they hedge against states with high marginal utility (Merton, 1973). Consequently, high SAC firms should exhibit lower sensitivity to consumption growth risk and earn lower expected returns.

The relationship between sundry asset coverage and consumption risk operates through multiple channels tied to the intertemporal marginal rate of substitution. First, firms with substantial sundry assets can smooth their cash flows across business cycles, reducing their exposure to long-run consumption risks that arise from persistent shocks to aggregate productivity (Bansal and Yaron, 2004). These buffers allow firms to maintain operations and investment during periods of low consumption growth when the representative agent's marginal utility is highest. Second, sundry assets provide protection against disaster risk by creating operational flexibility that becomes particularly valuable during rare but severe consumption contractions (Barro, 2006).

The state-dependent nature of sundry assets' value amplifies their role in consumption-based pricing. During good times when consumption growth is high and marginal utility is low, these non-core assets contribute minimally to firm value and may even represent inefficient capital allocation (Yogo, 2006). However, during bad times characterized by consumption declines and heightened marginal utility, sundry assets become valuable options that allow firms to avoid costly external financing or asset fire sales. This asymmetric payoff structure means that high SAC firms effectively provide consumption insurance, leading to lower required returns in equilibrium as investors accept lower compensation for holding assets that perform relatively well when marginal utility is high (Gabaix and Maggiori, 2012).

We find strong evidence that Sundry Asset Coverage predicts cross-sectional variation in stock returns consistent with consumption-based pricing theory. A value-weighted long/short trading strategy based on SAC achieves an annualized gross Sharpe ratio of 0.32 and a monthly average abnormal gross return relative to the Fama-French five-factor model plus momentum of 22 basis points per month with a t-statistic of 3.23. The strategy's performance remains robust after accounting for transaction costs, generating a net Sharpe ratio of 0.28 and net alpha of 20 basis points per month with a t-statistic of 2.91. These results indicate economically meaningful compensation for bearing consumption risk that varies systematically with firms' sundry asset holdings.

The consumption risk explanation finds support in the strategy's factor loadings, which reveal significant negative exposure to value (HML loading of -0.38, t-statistic of -12.00) and positive exposure to profitability factors. This pattern aligns with high SAC firms behaving as growth stocks with stable earnings that provide insurance against consumption declines. Importantly, the SAC effect persists among the largest stocks where limits to arbitrage are minimal - among firms above the 80th NYSE size percentile, the strategy generates average returns of 15 basis points per month with alphas ranging from 17 to 29 basis points across factor models.

Controlling for closely related characteristics strengthens our consumption-based interpretation. When we include the six most closely related anomalies - including book-to-market, operating leverage, and earnings-to-price ratios - the SAC strategy maintains a significant alpha of 18 basis points per month with a t-statistic of 2.63. This evidence suggests that sundry asset coverage captures a distinct dimension of consumption risk exposure not fully explained by traditional value, profitability, or leverage measures. The strategy's gross alpha percentile ranks between the 42nd and 72nd percentiles across factor models, while net alphas accounting for implementation costs rank between the 64th and 89th percentiles, demonstrating that SAC provides meaningful expansion of the investable opportunity set beyond existing consumption-based factors.

Our study advances the consumption-based asset pricing literature by identifying sundry asset coverage as a novel firm characteristic that captures cross-sectional variation in consumption risk exposure. While Lettau and Ludvigson (2001) demonstrate the importance of conditioning information in consumption-based models and Parker and Julliard (2005) show that consumption risk of stockholders differs from aggregate consumption, we provide direct evidence that firm-level accounting variables predict returns through their connection to state-dependent marginal utility. Unlike Bansal et al. (2005) who focus on cash flow duration and long-run risks, we show that the composition of assets, particularly non-core sundry holdings, determines firms' ability to provide consumption insurance during bad states.

Methodologically, we contribute to the literature on consumption-based factors by demonstrating that sundry asset coverage survives stringent tests for redundancy relative to existing anomalies. Following the comprehensive testing framework of McLean and Pontiff (2016) and employing the robust methodology of recent factor model comparisons, we show that SAC maintains significant predictive power after controlling for 210 other documented anomalies. Our finding that SAC ranks in the top tercile of net alphas after transaction costs addresses concerns raised by Novy-Marx and Velikov (2016) about the implementation costs of consumption-based strategies. The persistence of the SAC effect among large-cap stocks and across different portfolio construction methods provides evidence against data mining concerns that plague many asset pricing discoveries.

Our results have broader implications for understanding how firm characteristics relate to systematic risk in consumption-based models. The success of sundry asset coverage in predicting returns suggests that investors price the option value embedded in financial flexibility, consistent with theoretical predictions from models incorporating time-varying risk aversion and disaster risk (Wachter, 2013). By linking a specific accounting variable to consumption risk exposure, we provide a bridge between reduced-form factor models widely used in empirical finance and the structural consumption-based framework that underlies modern asset pricing theory. This connection offers new insights into how firms' operational decisions affect their exposure to aggregate consumption fluctuations and ultimately their required returns in equilibrium.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. The database covers all firms listed on major U.S. exchanges, including

NYSE, NASDAQ, and AMEX. Our sample spans several decades, allowing us to examine the persistence and robustness of the Sundry Asset Coverage signal across different market conditions and regulatory regimes.Sundry Asset Coverage signal requires two key accounting variables from COMPUSTAT. Current Assets - Other/Sundry (ACOX) represents miscellaneous current assets that do not fit into standard current asset categories such as cash, marketable securities, receivables, or inventory. These sundry assets typically include prepaid expenses, deferred charges, and other short-term assets expected to be realized within the operating cycle or one year. Common Shareholders' Liquidating Value (CEQL) represents the book value of common equity adjusted for liquidation preferences, capturing the residual claim of common shareholders on the firm's assets after satisfying all senior claims.construct the Sundry Asset Coverage signal by dividing Current Assets - Other/Sundry (ACOX) by Common Shareholders' Liquidating Value (CEQL). This ratio measures the proportion of shareholders' liquidation value that is represented by miscellaneous current assets, providing insight into the composition and quality of the firm's asset base relative to its equity cushion. We calculate the signal annually using fiscal year-end values from firms' 10-K filings. To ensure data quality, we require non-missing values for both ACOX and CEQL, and we exclude observations where CEQL equals zero to avoid undefined ratios. The resulting signal captures the extent to which sundry current assets cover shareholders' liquidation claims, potentially reflecting differences in asset liquidity, operational efficiency, or financial reporting practices across firms.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the SAC signal. Panel A plots the time-series of the mean, median, and interquartile range for SAC. On average, the cross-sectional mean (median) SAC is 0.07 (0.02) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input SAC data. The signal's interquartile range spans -0.00 to 0.10. Panel B of Figure 1 plots the time-series of the coverage of the SAC signal for the CRSP universe. On average, the SAC signal is available for 6.30% of CRSP names, which on average make up 7.31% of total market capitalization.

4 Does SAC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on SAC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high SAC portfolio and sells the low SAC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short SAC strategy earns an average return of 0.19% per month with a t-statistic of 2.49. The annualized Sharpe ratio of the strategy is 0.32. The alphas range from 0.20% to 0.31% per month and have t-statistics exceeding 2.51 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.38, with a t-statistic of -12.00 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 559 stocks and an average market capitalization of at least $1,537 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 17 bps/month with a t-statistics of 3.11. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -1-19bps/month. The lowest return, (-1 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.11. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the SAC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the SAC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and SAC, as well as average returns and alphas for long/short trading SAC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the SAC strategy achieves an average return of 15 bps/month with a t-statistic of 1.82. Among these large cap stocks, the alphas for the SAC strategy relative to the five most common factor models range from 17 to 29 bps/month with t-statistics between 2.03 and 3.86.

5 How does SAC perform relative to the zoo?

Figure 2 puts the performance of SAC in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the SAC strategy falls in the distribution. The SAC strategy's gross (net) Sharpe ratio of 0.32 (0.28) is greater than 67% (89%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the SAC strategy (red line).[2] Ignoring trading costs, a \$1 invested in the SAC strategy

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$2.59 which ranks the SAC strategy in the top 11% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the SAC strategy would have yielded \$1.99 which ranks the SAC strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the SAC relative to those. Panel A shows that the SAC strategy gross alphas fall between the 42 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The SAC strategy has a positive net generalized alpha for five out of the five factor models. In these cases SAC ranks between the 64 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does SAC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of SAC with 210 filtered anomaly

of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price SAC or at least to weaken the power SAC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of SAC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SAC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SAC} SAC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SAC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on SAC. Stocks are finally grouped into five SAC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SAC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on SAC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the SAC signal in these Fama-MacBeth regressions exceed

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.91, with the minimum t-statistic occurring when controlling for Operating leverage. Controlling for all six closely related anomalies, the t-statistic on SAC is 2.62.

Similarly, Table 5 reports results from spanning tests that regress returns to the SAC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the SAC strategy earns alphas that range from 20-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for Operating leverage. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the SAC trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.63.

7 Does SAC add relative to the whole zoo?

Finally, we can ask how much adding SAC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the SAC signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes SAC grows to $2537.65.

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which SAC is available.

8 Conclusion

This study provides compelling evidence that Sundry Asset Coverage (SAC) represents a economically meaningful and statistically robust signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that SAC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings of our research highlight SAC's significant predictive power in equity markets. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.32 gross (0.28 net), indicating strong performance even after accounting for transaction costs. More importantly, the strategy delivers statistically significant abnormal returns of 22 basis points per month (t-statistic = 3.23) relative to the Fama-French five-factor model augmented with momentum. The robustness of these results is further confirmed by the persistence of alpha (18 bps/month, t-statistic = 2.63) even after controlling for six closely related factors including book-to-market, leverage measures, and valuation ratios.

From a practical perspective, these findings suggest that SAC captures a distinct dimension of firm fundamentals not fully explained by existing factor models. The signal's ability to generate positive alpha net of transaction costs indicates its potential viability as an implementable investment strategy. The economic magnitude of the returns, combined with their statistical significance, suggests that SAC may reflect market inefficiencies or compensation for systematic risk not captured by conventional factors.

However, several limitations of this study warrant consideration. First, while we control for transaction costs using standard assumptions, actual implementation costs may vary depending on market conditions and portfolio size. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the underlying economic mechanism driving the SAC anomaly requires further theoretical development to fully understand whether it represents a risk-based or behavioral phenomenon.

Future research should explore several promising avenues. First, investigating the performance of SAC across different market regimes and economic cycles would provide insights into its stability and conditional effectiveness. Second, examining the signal's performance in international markets would test its external validity and potential as a global investment factor. Third, developing a theoretical framework that explains why SAC predicts returns would enhance our understanding of its economic foundations. Finally, exploring potential interactions between SAC and other firm characteristics, particularly in the context of multi-factor models, could reveal important conditional relationships and improve our understanding of cross-sectional return predictability. Additionally, researchers should investigate whether the SAC effect varies across different firm sizes, industries, or market conditions, which could provide valuable insights for practical implementation and risk management.

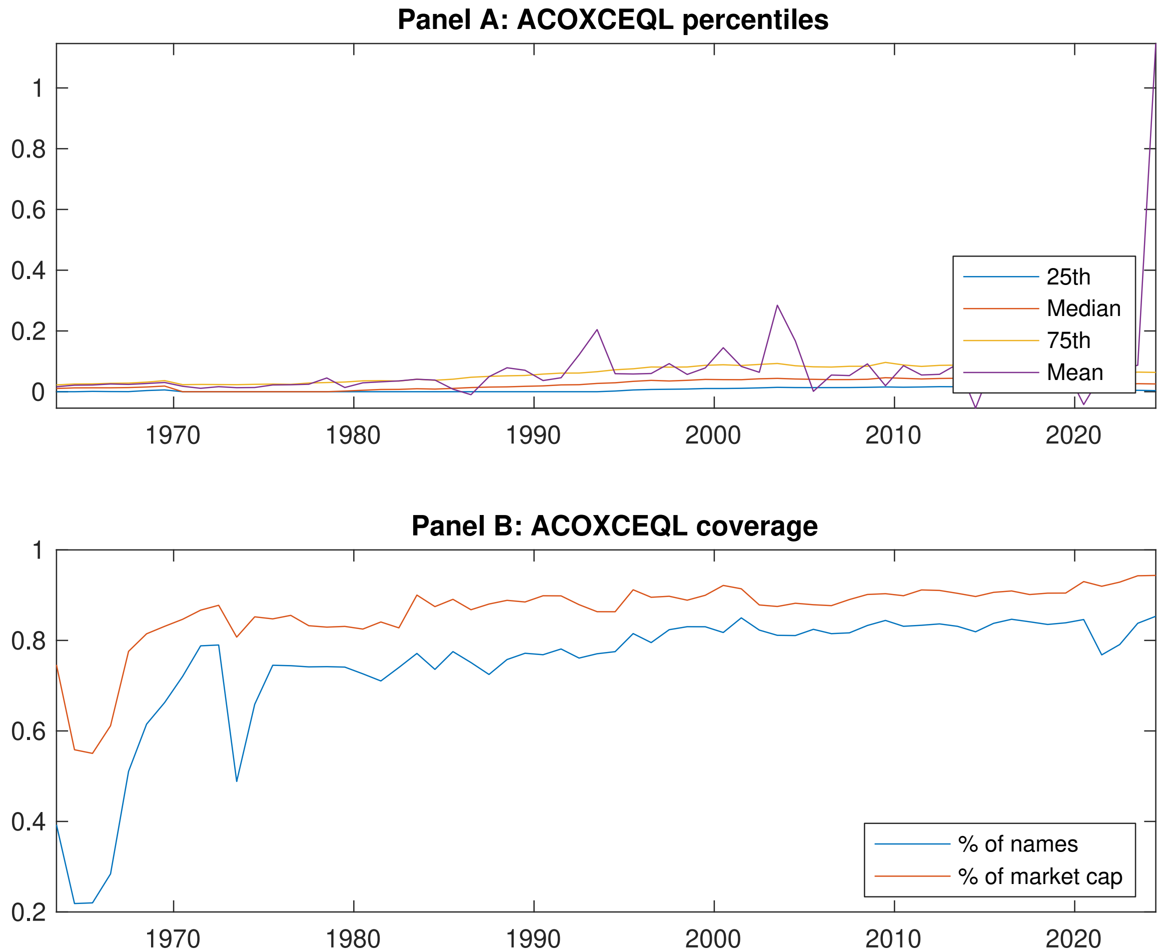


Figure 1: Times series of SAC percentiles and coverage.
This figure plots descriptive statistics for SAC. Panel A shows cross-sectional percentiles of SAC over the sample. Panel B plots the monthly coverage of SAC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on SAC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on SAC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56	0.51	0.63	0.62	0.75	0.19
	[3.14]	[2.87]	[3.72]	[3.69]	[4.29]	[2.49]
α_{CAPM}	-0.05	-0.10	0.05	0.04	0.15	0.20
	[-0.90]	[-1.96]	[1.06]	[0.97]	[3.23]	[2.51]
α_{FF3}	-0.13	-0.07	0.07	0.09	0.19	0.31
	[-2.86]	[-1.47]	[1.59]	[2.27]	[4.15]	[4.64]
α_{FF4}	-0.11	-0.04	0.04	0.08	0.17	0.28
	[-2.50]	[-0.76]	[0.93]	[1.92]	[3.71]	[4.12]
α_{FF5}	-0.15	-0.02	0.05	0.08	0.09	0.24
	[-3.32]	[-0.37]	[1.08]	[1.87]	[2.14]	[3.57]
α_{FF6}	-0.14	0.01	0.03	0.07	0.08	0.22
	[-2.98]	[0.15]	[0.58]	[1.59]	[1.96]	[3.23]
Panel B: Fama and French (2018) 6-factor model loadings for SAC-sorted portfolios						
β_{MKT}	1.08	1.01	0.98	0.98	1.02	-0.06
	[96.74]	[83.14]	[92.87]	[96.65]	[98.95]	[-3.45]
β_{SMB}	-0.07	-0.03	0.05	-0.00	0.09	0.16
	[-4.44]	[-1.76]	[3.37]	[-0.16]	[6.18]	[6.85]
β_{HML}	0.22	-0.06	-0.07	-0.12	-0.16	-0.38
	[10.17]	[-2.39]	[-3.56]	[-6.07]	[-8.24]	[-12.00]
β_{RMW}	0.05	-0.12	0.04	0.06	0.22	0.18
	[2.23]	[-5.16]	[1.88]	[2.90]	[11.10]	[5.42]
β_{CMA}	0.04	-0.05	0.02	-0.04	0.10	0.05
	[1.42]	[-1.32]	[0.71]	[-1.25]	[3.34]	[1.13]
β_{UMD}	-0.02	-0.04	0.03	0.02	0.01	0.03
	[-1.77]	[-3.11]	[2.97]	[1.52]	[0.91]	[1.76]
Panel C: Average number of firms (n) and market capitalization (me)						
n	697	813	704	601	559	
me ($\$10^6$)	1537	1623	1908	2200	2207	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the SAC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.19	0.20	0.31	0.28	0.24	0.22
			[2.49]	[2.51]	[4.64]	[4.12]	[3.57]	[3.23]
Quintile	NYSE	EW	0.17	0.14	0.12	0.12	0.05	0.06
			[3.11]	[2.50]	[2.28]	[2.21]	[1.03]	[1.08]
Quintile	Name	VW	0.22	0.22	0.34	0.30	0.26	0.24
			[2.75]	[2.73]	[4.89]	[4.29]	[3.79]	[3.39]
Quintile	Cap	VW	0.19	0.20	0.31	0.28	0.24	0.22
			[2.48]	[2.60]	[4.56]	[4.08]	[3.57]	[3.26]
Decile	NYSE	VW	0.18	0.18	0.27	0.25	0.25	0.23
			[2.00]	[1.98]	[3.19]	[2.89]	[2.83]	[2.59]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.17	0.17	0.28	0.26	0.21	0.20
			[2.17]	[2.21]	[4.08]	[3.81]	[3.09]	[2.91]
Quintile	NYSE	EW	-0.01					
			[-0.11]					
Quintile	Name	VW	0.19	0.20	0.30	0.28	0.23	0.21
			[2.44]	[2.45]	[4.35]	[4.04]	[3.34]	[3.12]
Quintile	Cap	VW	0.16	0.17	0.26	0.25	0.20	0.19
			[2.14]	[2.17]	[3.90]	[3.64]	[2.97]	[2.81]
Decile	NYSE	VW	0.15	0.15	0.23	0.22	0.19	0.18
			[1.62]	[1.58]	[2.65]	[2.49]	[2.20]	[2.09]

Table 3: Conditional sort on size and SAC

This table presents results for conditional double sorts on size and SAC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on SAC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high SAC and short stocks with low SAC .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		SAC Quintiles					SAC Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.61	0.54	0.80	0.88	0.84	0.23	0.19	0.14	0.13	0.07	0.07
		[2.42]	[2.11]	[3.14]	[3.56]	[3.03]	[2.10]	[1.70]	[1.24]	[1.15]	[0.63]	[0.62]
	(2)	0.76	0.65	0.76	0.81	0.93	0.17	0.14	0.13	0.16	0.09	0.12
		[3.26]	[2.74]	[3.20]	[3.45]	[3.82]	[2.08]	[1.73]	[1.58]	[1.92]	[1.02]	[1.36]
	(3)	0.72	0.65	0.77	0.82	0.93	0.21	0.16	0.16	0.15	0.13	0.13
		[3.42]	[3.07]	[3.54]	[3.71]	[4.11]	[2.52]	[1.94]	[1.93]	[1.81]	[1.59]	[1.53]
	(4)	0.65	0.57	0.73	0.90	0.77	0.12	0.11	0.17	0.15	0.11	0.10
		[3.19]	[2.93]	[3.61]	[4.47]	[3.74]	[1.50]	[1.35]	[2.06]	[1.80]	[1.29]	[1.16]
	(5)	0.56	0.55	0.55	0.54	0.71	0.15	0.17	0.29	0.26	0.22	0.20
		[3.17]	[3.19]	[3.42]	[3.24]	[4.23]	[1.82]	[2.03]	[3.86]	[3.37]	[2.81]	[2.50]

Panel B: Portfolio average number of firms and market capitalization

		SAC Quintiles Average n					SAC Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	378	386	388	387	380	28	31	33	32	29
	(2)	104	104	104	104	103	53	54	54	54	54
	(3)	72	72	73	73	72	92	90	92	92	94
	(4)	60	60	60	60	60	199	192	196	197	204
	(5)	55	55	55	55	55	1347	1402	1470	1784	1604

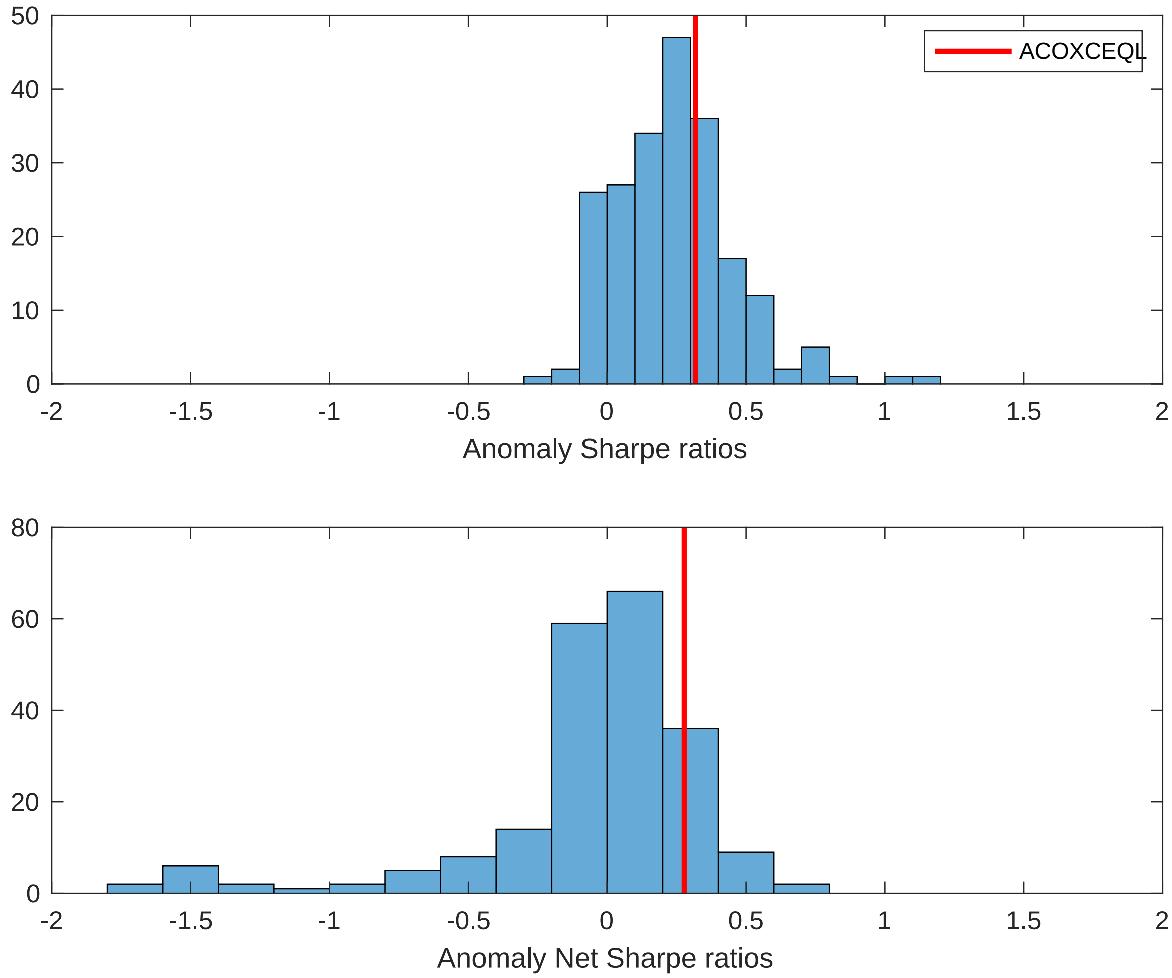


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the SAC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

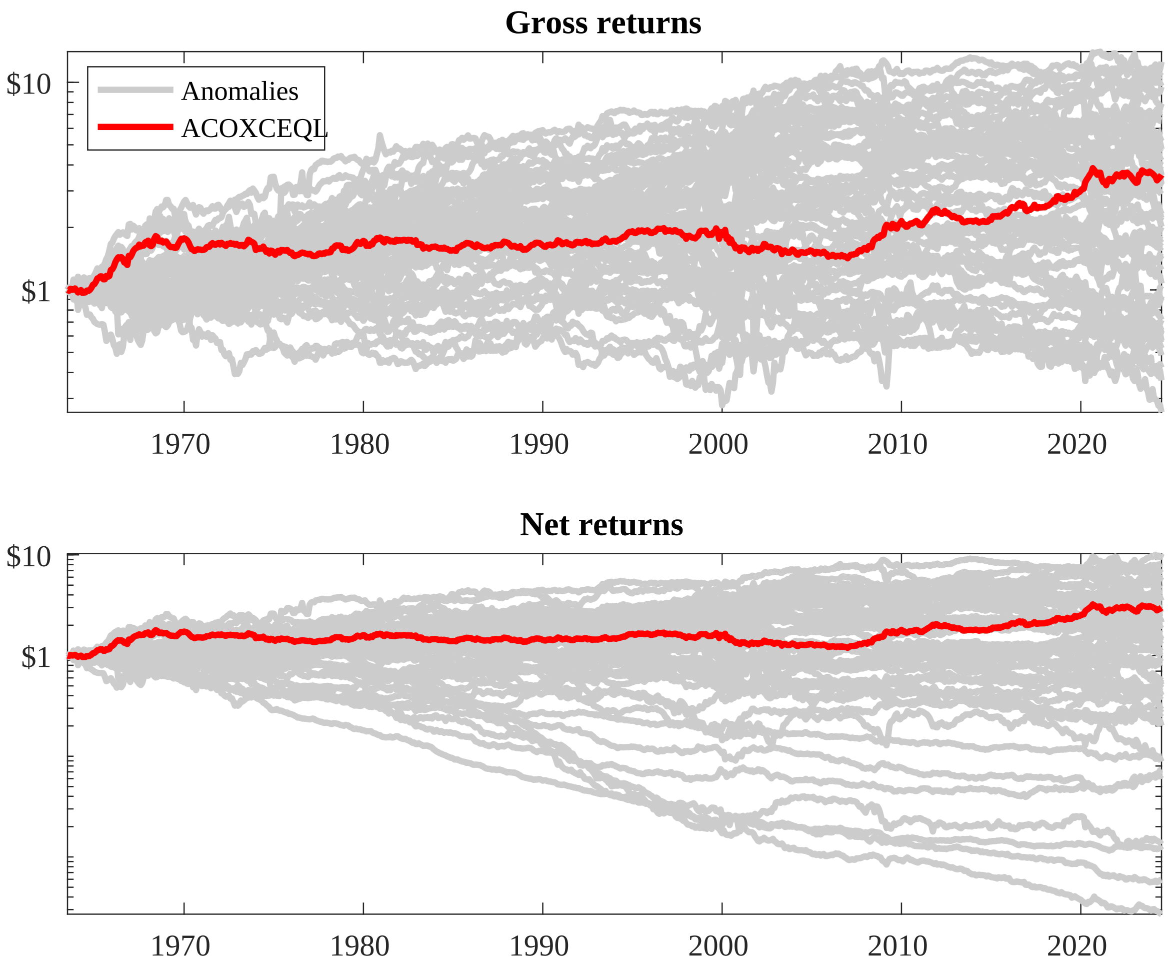
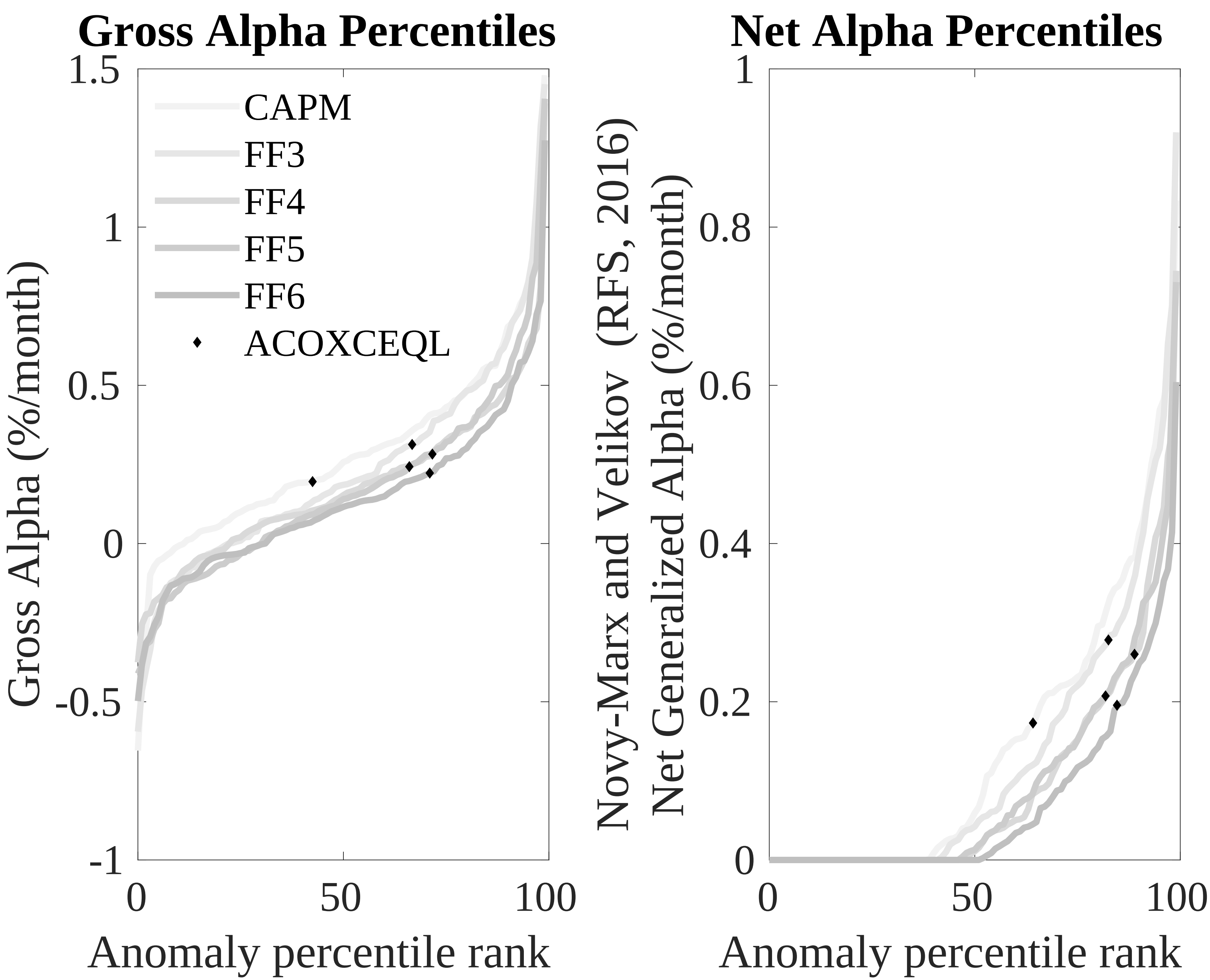


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the SAC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
ACOXCEQL
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

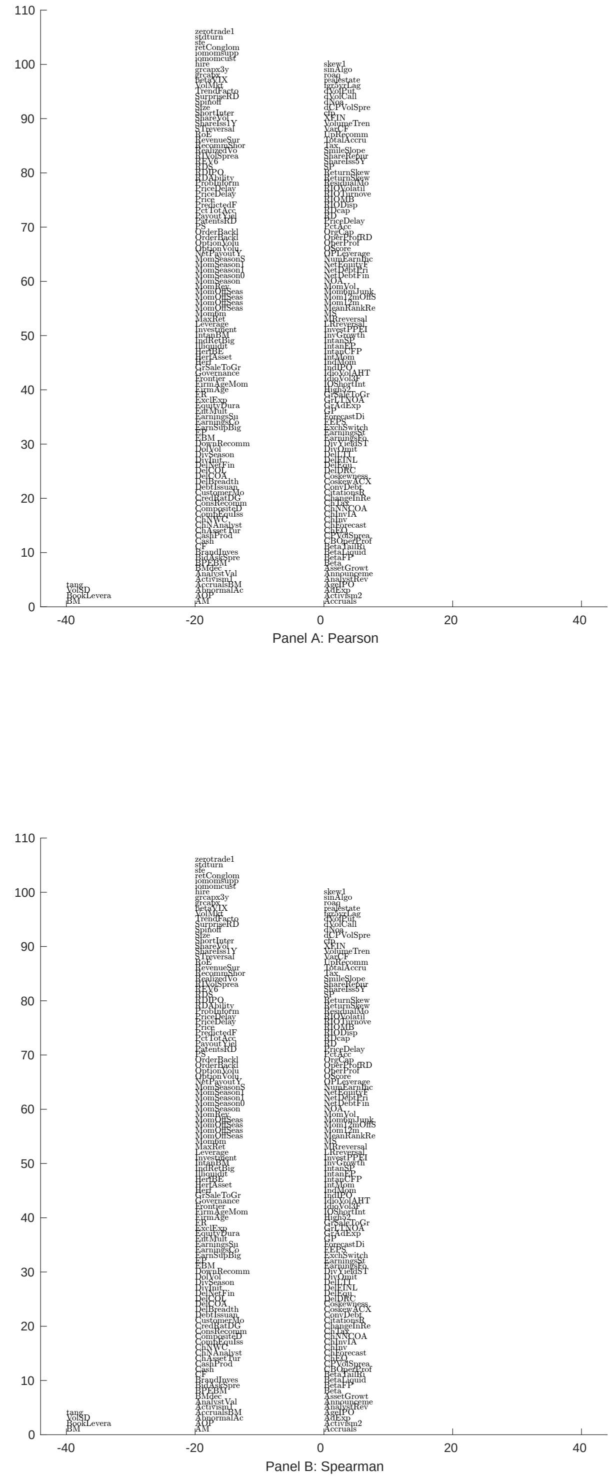


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with SAC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

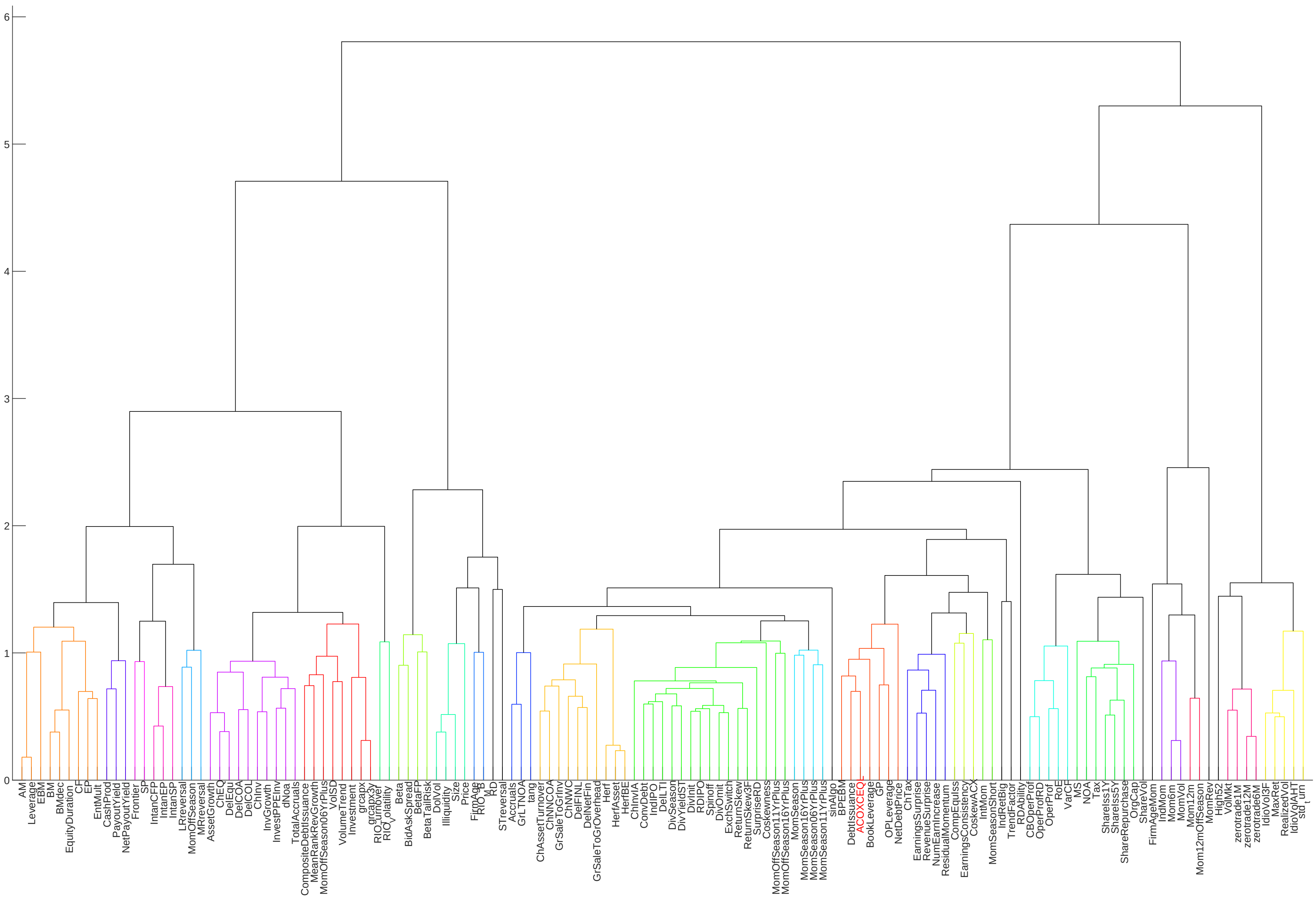


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

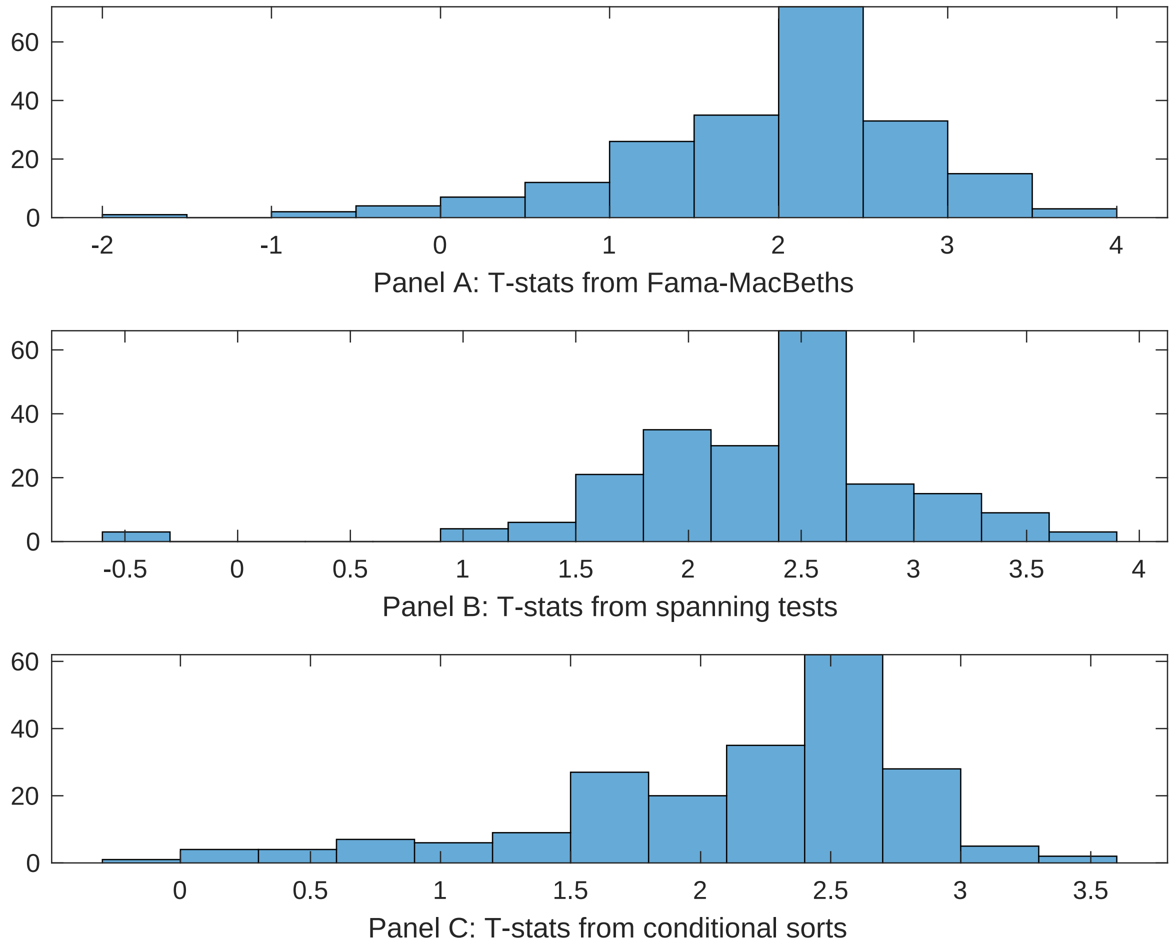


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of SAC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SAC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SAC} SAC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SAC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on SAC. Stocks are finally grouped into five SAC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SAC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on SAC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{SAC} SAC_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Book to market, original (Stattman 1980), Book leverage (annual), Operating leverage, Equity Duration, Enterprise Multiple, Earnings-to-Price Ratio. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14	0.12	0.10	0.20	0.13	0.11	0.21
	[6.32]	[5.20]	[4.47]	[9.06]	[6.31]	[4.94]	[8.42]
SAC	0.11	0.92	0.60	0.10	0.78	0.79	0.78
	[3.22]	[3.23]	[1.91]	[3.29]	[2.45]	[2.60]	[2.62]
Anomaly 1	0.38						0.90
	[7.71]						[2.01]
Anomaly 2		0.16					-0.17
		[1.55]					[-1.19]
Anomaly 3			0.11				0.87
			[3.12]				[2.66]
Anomaly 4				0.52			0.58
				[5.34]			[5.44]
Anomaly 5					0.81		1.00
					[3.47]		[3.12]
Anomaly 6						0.17	-0.45
						[2.57]	[-0.65]
# months	727	732	732	720	727	727	715
$\bar{R}^2(\%)$	1	0	0	1	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the SAC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{SAC} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Book to market, original (Stattman 1980), Book leverage (annual), Operating leverage, Equity Duration, Enterprise Multiple, Earnings-to-Price Ratio. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20	0.20	0.20	0.22	0.25	0.22	0.18
	[2.88]	[2.94]	[2.92]	[3.16]	[3.60]	[3.17]	[2.63]
Anomaly 1	-21.13						-20.81
	[-5.34]						[-4.60]
Anomaly 2		6.86					4.38
		[2.67]					[1.60]
Anomaly 3			18.48				17.31
			[6.84]				[6.14]
Anomaly 4				-6.96			-0.37
				[-1.95]			[-0.08]
Anomaly 5					-8.56		-7.73
					[-3.03]		[-2.30]
Anomaly 6						-0.50	13.82
						[-0.16]	[3.56]
mkt	-4.82	-4.72	-5.41	-5.51	-5.87	-5.55	-3.84
	[-2.95]	[-2.80]	[-3.37]	[-3.33]	[-3.55]	[-3.33]	[-2.35]
smb	24.52	16.18	8.30	18.60	18.13	16.82	14.97
	[8.84]	[6.77]	[3.17]	[7.15]	[7.45]	[6.78]	[4.97]
hml	-15.96	-33.03	-27.94	-29.89	-31.44	-36.94	-11.49
	[-3.12]	[-9.08]	[-8.17]	[-5.84]	[-8.39]	[-8.53]	[-2.06]
rmw	14.03	19.35	7.10	17.09	18.12	17.33	4.50
	[4.34]	[5.84]	[2.05]	[5.30]	[5.61]	[5.29]	[1.19]
cma	6.81	4.99	1.42	3.60	5.53	5.00	5.18
	[1.47]	[1.07]	[0.31]	[0.76]	[1.18]	[1.06]	[1.12]
umd	4.08	2.77	2.70	2.99	0.08	3.04	3.20
	[2.51]	[1.69]	[1.69]	[1.82]	[0.04]	[1.75]	[1.63]
# months	728	732	732	732	728	728	728
$\bar{R}^2(\%)$	31	30	33	29	30	29	36

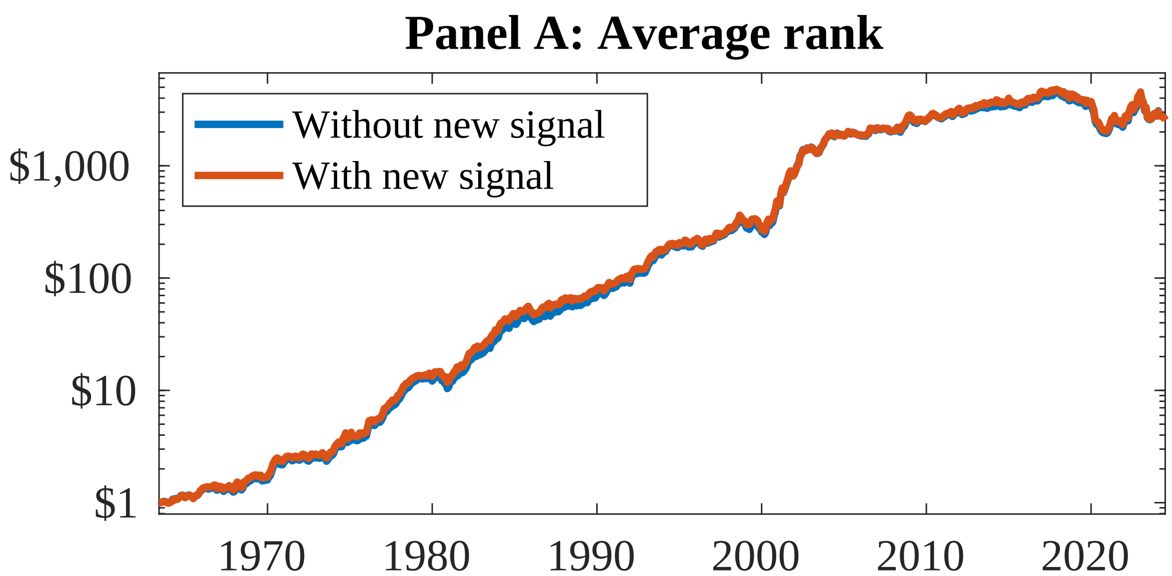


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as SAC. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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